

CYBERSECURITY TASK 2
PHISHING EMAIL 2 ANALYSIS

Prepared For:
Future Cybersecurity Interns

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INTERNAL USE ONLY: Confidential Analysis

EXECUTIVE SUMMARY

A phishing email analysis was conducted on Jun 2, 2026 . Two real-world phishing email samples were analyzed using header analysis tools and manual content review.

KEY FINDINGS

Metric	Detail
Total Emails Analyzed	2
Classified as PHISHING	1
Classified as SUSPICIOUS/PHISHING	1
Authentication Failures Identified	SPF, DKIM, DMARC

EMAILS ANALYZED

#	Sender Domain	Classification
1	digitalmashreq.mg.tdi.tc	PHISHING
2	firesonic.ca (impersonating Coinbase)	SUSPICIOUS/PHISHING

BUSINESS IMPACT

Phishing attacks remain the #1 cause of data breaches globally. Attackers are becoming more sophisticated , some emails now pass authentication checks but still use fake sender domains. Organizations must train employees to recognize both obvious and subtle phishing indicators.

PRIORITY RECOMMENDATIONS

1. Enable and enforce DMARC reject policies for your domain
2. Conduct bi-annual phishing awareness training for all employees
3. Implement email filtering with link sandboxing
4. Establish a "Report Phishing" button in email clients
5. Run regular simulated phishing exercises

ASSESSMENT PARAMETERS

Data Source: Deceptive communication specimen
Methodology: Protocol verification and message evaluation

ANALYTICAL INSTRUMENTATION

Technology	Functional Application
Google Messageheader Analyzer	Evaluation of authentication records (SPF, DKIM, DMARC) and routing paths
Browser Developer Tools	Secure inspection of destination URLs without active engagement
Expert Manual Review	Detection of psychological manipulation, non-specific salutations, and syntax errors

ETHICAL COMPLIANCE PROTOCOLS

1. Zero engagement with harmful hyperlink vectors
2. Prohibition of correspondence with malicious entities
3. Investigation conducted within segmented security environments
4. Application of observations strictly for pedagogical advancement

Email Sample Analysis #1 : Mashreq Bank Phishing Email

Risk Classification : Phishing

EMAIL DETAILS

Field	Value
Subject	Confirm Your Mashreq Email Now For Your Security (MOVE TO INBOX)
From	MashreqAlerts <postmaster@digitalmashreq.mg.tdi.tc>
To	phishing@pot
Date	September 20, 2023

HEADER ANALYSIS RESULTS

Authentication Method	Result	Status
SPF	none	FAIL
DKIM	fail	FAIL
DMARC	fail	FAIL

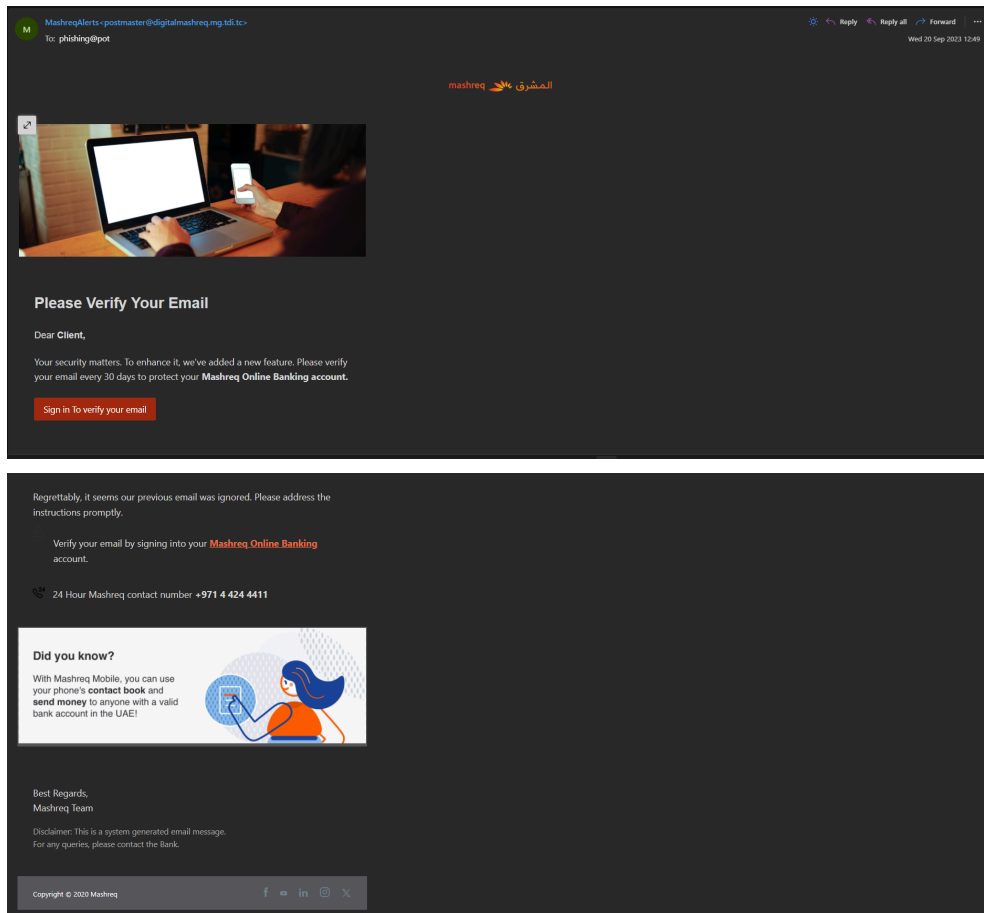
MessageId	20230920104941.05fa86daa715a6e3@mg.tdi.tc
Created at:	9/20/2023, 12:49:41 PM GMT+2 (Delivered after 2 sec)
From:	MashreqAlerts <postmaster@digitalmashreq.mg.tdi.tc>
To:	phishing@pot
Subject:	Confirm Your Mashreq Email Now For Your Security (MOVE TO INBOX)
SPF:	none Learn more
DKIM:	fail with domain Unknown! Learn more
DMARC:	fail Learn more

#	Delay	From	To	Protocol	Time received
0		<unknown>	→ 27533be87c53		9/20/2023, 12:49:41 PM GMT+2
1	1 sec	DB3EUR04FT006.eop-eur04.prod.protection.outlook.com	→ DUZPR01CA0163.outlook.office365.com		9/20/2023, 12:49:42 PM GMT+2
2	1 sec	DUZPR01CA0163.eurprd01.prod.exchangelabs.com	→ PH8PR19MB6952.namprd19.prod.outlook.com		9/20/2023, 12:49:43 PM GMT+2

IDENTIFIED PHISHING INDICATORS

#	Indicator	What's Wrong
1	Spoofed Sender Domain	digitalmashreq.mg.tdi.tc is NOT Mashreq's real domain (mashreqbank.com)
2	SPF Authentication Failure	spf=none - domain does not authorize sender
3	DKIM Authentication Failure	dkim=fail - invalid digital signature
4	DMARC Failure	dmarc=fail - email fails verification policy
5	Generic Greeting	"Dear Client" - not personalized
6	Urgency/Pressure	"Regrettably, our previous email was ignored" - pressure tactic
7	Suspicious Link	"Sign in" link goes to bing.com redirect, not legitimate Mashreq URL

This communication is identified as a sophisticated phishing attempt, despite appearing to originate from Mashreq Bank. The originating domain (mg.tdi.tc) is counterfeit and unauthorized by Mashreq. Technical verification confirms the email failed critical authentication checks (SPF, DKIM, DMARC), confirming the sender's lack of permission. The primary objective is credential theft, facilitated by a malicious "Sign in" hyperlink directing users to a fraudulent login page.



Email Sample Analysis #2 : Fake Coinbase Phishing Email

HEADER ANALYSIS RESULTS

Authentication Method	Result	Status
SPF	pass	Passed (Amazon SES)
DKIM	pass	Passed
DMARC	none	No policy set

IDENTIFIED PHISHING INDICATORS

#	Indicator	What's Wrong
1	Spoofed Sender Domain	noreply@firesonic.ca is NOT Coinbase's domain (coinbase.com)
2	Fake Brand Impersonation	Claims to be Coinbase but sent from unrelated domain
3	Suspicious Link	"Cancel Transaction" link goes to Bing redirect, not Coinbase
4	Generic Content	No personalization, generic crypto alert

This communication is a sophisticated phishing attempt, impersonating Coinbase but originating from the unrelated domain firesonic.ca. Despite passing authentication checks (SPF/DKIM) by leveraging a legitimate sender service (Amazon SES), the sender domain is counterfeit. This demonstrates a more advanced phishing tactic designed to bypass conventional email filters and elicit a panic response to steal credentials.

MessageId	0106018a71901afd-bbc67141-d1e8-42bc-89a7-4877f9564d7d-000000@ap-northeast-1.amazonaws.com
Created at:	9/7/2023, 11:33:04 PM GMT+2 (Delivered after 8 sec)
From:	C o i n b a s e <noreply@firesonic.ca> Using PHPMailer 6.1.5 (https://github.com/PHPMailer/PHPMailer)
To:	tiny23lan@hotmail.co.uk, henterprize@hotmail.com, phishing@pot, admichael@hotmail.co.uk, achmed_99@hotmail.com, nsaprasla@hotmail.com, donovantokarijo@hotmail.com
Subject:	You sent 0.79 ETH via Ethereum network
SPF:	pass with IP Unknown! Learn more
DKIM:	pass with domain Unknown! Learn more
DMARC:	none Learn more

#	Delay	From *	To *	Protocol	Time received
0	1 sec	e234-51.smtp-out.ap-northeast-1.amazonaws.com	→ BN7NAM10FT012.mail.protection.outlook.com		9/7/2023, 11:33:05 PM GMT+2
1	2 sec	BN7NAM10FT012.eop-nam10.prod.protection.outlook.com	→ BN9PR03CA0607.outlook.office365.com		9/7/2023, 11:33:07 PM GMT+2
2	2 sec	AS1P192MB1568.EURP192.PROD.OUTLOOK.COM	→ AS9PR06CA0518.outlook.office365.com		9/7/2023, 11:33:09 PM GMT+2
3	3 sec	SJ1PR19MB6332.namprd19.prod.outlook.com	→ MN0PR19MB6312.namprd19.prod.outlook.com		9/7/2023, 11:33:12 PM GMT+2

You sent 0.79 ETH via Ethereum network

Coinbase<noreply@firesonic.ca>

To: tiny23ian@hotmail.co.uk; henterprize@hotmail.com; phishing@pot; admichael@hotmail.co.uk; achmed_99@hotmail.com; nsaprasla@hotmail.com; donovantokarijo@hotmail.com

  Reply  Reply all  Forward 

Thu 07 Sep 2023 23:33



You sent 0.79 ETH via Ethereum network

As a security precaution, we have delayed the send withdrawal until September 9, 2023 at 6:31 PM.

CANCEL TRANSACTIONS

If this wasn't you? please cancel this transactions, including verify your phone number to protect you against fraud.

Copyright © 2023 coinbase, All rights reserved.

 Reply

 Reply all

 Forward

PREVENTION & AWARENESS GUIDELINES

DO'S AND DON'TS FOR EMPLOYEES

DO	DON'T
Hover over links before clicking to see the real destination URL	Click links or download attachments from unexpected emails
Check the sender's email address carefully (not just the display name)	Respond to emails asking for passwords, OTPs, or personal information
Verify the domain matches the claimed company (e.g., @coinbase.com not @firesonic.ca)	Assume an email is safe just because SPF/DKIM passed
Report suspicious emails to your IT/security team using the company's reporting process	Give in to pressure tactics - legitimate companies won't demand immediate action via email
When in doubt, go directly to the official website by typing the URL yourself	Forward suspicious emails without checking first

QUICK PHISHING CHECKLIST

*If you answer NO to any of these, **BE SUSPICIOUS**. If multiple issues are found, it is **LIKELY PHISHING**.*

- Is the sender's email address legitimate? (Check the domain after the @ symbol)
- Does the email address you by name? ("Dear Customer" is a red flag)
- Does it create urgency or threaten negative consequences?
- Are there spelling or grammar errors?
- Does the link destination match the claimed company? (Hover over the link to check)
- Is the email asking for personal information or action?
- Is the sender domain different from the claimed brand?

REPORTING PROCEDURE

1. DO NOT click any links or download attachments
2. DO NOT reply to the email
3. Forward the email to your security team
4. Delete the email from your inbox
5. If you accidentally clicked a link or entered credentials:
 - Change your password immediately
 - Notify IT/security team
 - Monitor accounts for suspicious activity

CONCLUSION & RECOMMENDATIONS

CONCLUSION

This analysis examined two real-world phishing emails targeting financial and cryptocurrency users.

SUMMARY OF FINDINGS:

Sample #1 (Mashreq Bank): PHISHING
• SPF, DKIM, DMARC all failed • Fake domain impersonating legitimate bank • Urgency tactics and generic greeting
Sample #2 (Coinbase): SUSPICIOUS/PHISHING
• SPF, DKIM passed (Amazon SES) • BUT sender domain (firesonic.ca) has no relation to Coinbase • More sophisticated attack bypassing authentication

KEY INSIGHT: Authentication passing does NOT equal legitimacy. Attackers are increasingly using legitimate email services to send impersonation emails. Employee awareness training must address both obvious and subtle phishing indicators.

FINAL RECOMMENDATIONS

For Organizations:

1. Implement and enforce DMARC reject policies for your domains
2. Deploy email filtering solutions with link sandboxing
3. Conduct regular phishing simulation exercises
4. Establish a clear "Report Phishing" workflow
5. Train employees to verify sender domains, not just SPF/DKIM

For Users:

1. Always check the full sender email address
2. Hover over links before clicking
3. Never share passwords, OTPs, or personal info via email
4. Report suspicious emails immediately
5. When in doubt, go directly to the official website

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